RETAIL DATA AGENT · LEARNING LOOP · AS OF 2026-11-01

DESIGN ONLY · NOT BUILT

How does the agent remember that you like tables?

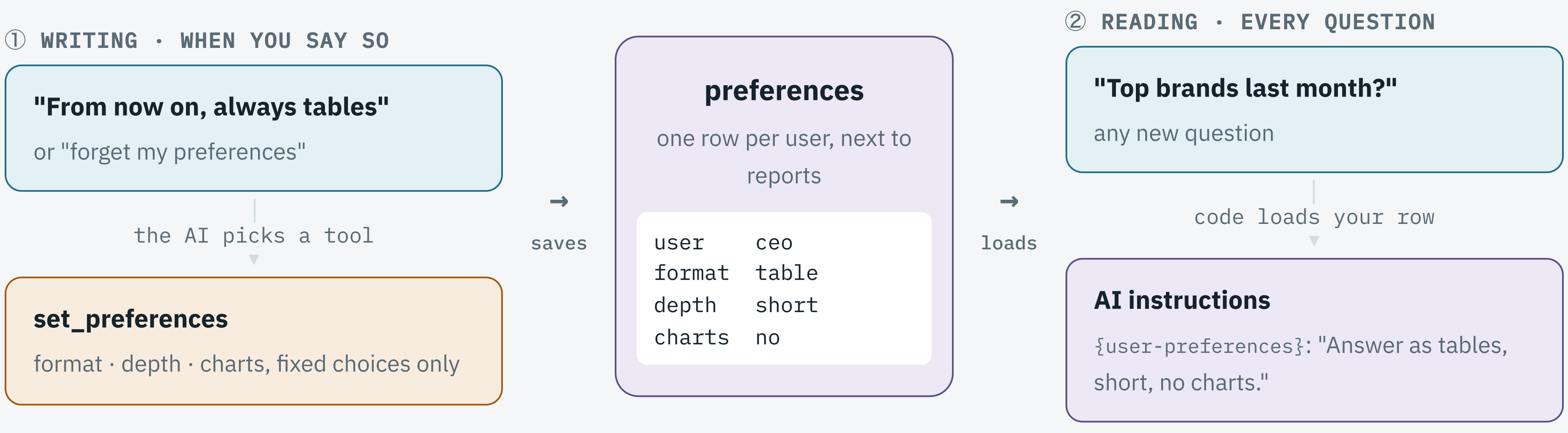
Each user gets one small row of settings, saved next to their reports. You change it by just saying so. Before every answer the agent reads that row, so the CEO gets short tables while a brand manager gets bullet points, from the same agent. See also Golden Knowledge and saved reports.

🟦 what the user says or sees 🟧 a tool writes 🟪 store and prompt 🟩 allowed 🟥 blocked

In short

1. **One row per user** in the same database as saved reports: format, depth, charts. Tick-boxes, not free text.
2. **Say it to change it.** "From now on, tables" calls a tool that saves the choice; "forget my preferences" clears it.
3. **Read before every answer.** The row is pasted into the AI's instructions, so the very next answer already follows it, with no extra AI call.
4. **Only looks, never access.** A preference can change how an answer is laid out, never which data is allowed.

The picture: one store, two lanes



① WRITING · WHEN YOU SAY SO

"From now on, always tables"
or "forget my preferences"

`the AI picks a tool`

set_preferences
format · depth · charts, fixed choices only

→ saves

preferences
one row per user, next to reports

```
user    ceo
format  table
depth   short
charts  no
```

→ loads

② READING · EVERY QUESTION

"Top brands last month?"
any new question

`code loads your row`

AI instructions
`{user-preferences}`: "Answer as tables, short, no charts."

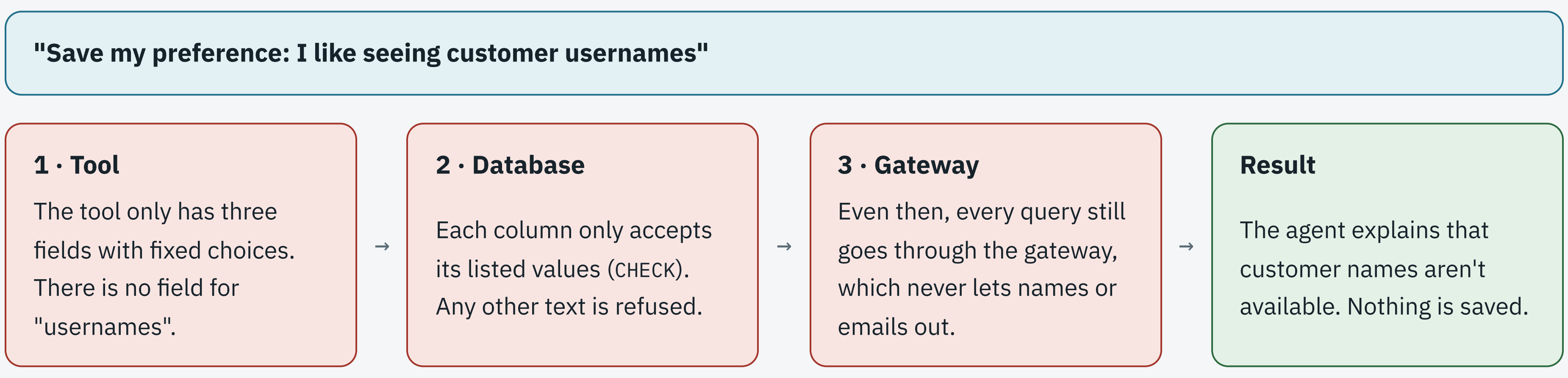
Learning over time (nightly)

A job reads the day's chats. If you asked "as a table" after three bullet answers, it saves a *suggestion*. Next time the agent asks once: "You often ask for tables. Make that your default?" Only your yes turns it on.

🏨 Like a hotel's guest card: check-in reads it ("quiet room, no feather pillows"), and the staff update it when you mention something new.

Why "I like seeing usernames" goes nowhere

A preference sits inside the AI's instructions, so a sneaky one could act like a hidden order. Three locks stop that, and each would stop it on its own:



"Save my preference: I like seeing customer usernames"

1 · Tool
The tool only has three fields with fixed choices. There is no field for "usernames".

→

2 · Database
Each column only accepts its listed values (`CHECK`). Any other text is refused.

→

3 · Gateway
Even then, every query still goes through the gateway, which never lets names or emails out.

→

Result
The agent explains that customer names aren't available. Nothing is saved.

And **whose** row gets written comes from the sign-in, not from the AI. The tools have no "user" input: the code fills it in, exactly like the owner of a saved report. So "change the CEO's preferences" from a brand manager can only ever touch the brand manager's own row.

Example: the CEO, over two days

CEO · MONDAY
From now on, give me short answers as tables.

AGENT
`set_preferences(format="table", depth="short")`
Saved: short answers, as tables.
(1 AI call: the tool's reply is the answer)

CEO · TUESDAY, NEW SESSION
Top 5 brands by revenue last month?

AGENT
A 5-row table, no long prose.
The row was read before the first AI call; nothing was asked again.

CEO
What do you remember about me?

AGENT
You prefer short answers as tables, without charts.
(answered from the instructions, no tool)

The two tools

TOOL	WHAT IT DOES	CONFIRMATION?
`set_preferences`	Saves one or more of `format` (table · bullets · text), `depth` (short · detailed), `charts` (yes · no) for the signed-in user.	No. It's low stakes; the agent just says what it saved.
`delete_preferences`	Clears one field, or all of them.	No. Nothing valuable is lost, unlike reports, which are hard-deleted only after a strict yes.
(no read tool)	The row is already in the instructions on every question.	—

```
CREATE TABLE preferences (
  user_id    TEXT PRIMARY KEY,                               # from sign-in, set by code
  format     TEXT CHECK (format IN ('table','bullets','text')),
  depth      TEXT CHECK (depth  IN ('short','detailed')),
  charts     TEXT CHECK (charts IN ('yes','no')),
  source     TEXT CHECK (source IN ('explicit','learned')),     # said by the user, or suggested nightly
  status     TEXT CHECK (status IN ('active','suggested')),     # a suggestion waits for the user's yes
  updated_at TEXT NOT NULL
)
```

> ### The rule in one sentence
>
> A preference is a tick-box the signed-in user sets in plain words; it is read before every answer and can change how the answer looks, never what data it may contain.

Status: design only (brief requirement 4a). Design: "User preferences" in `docs/hld.md`.

Would plug in as: a `preferences` table in the app database (the SQLite report store at `REPORTS_PATH` in the prototype, Cloud SQL in production); two tools made by factories that bind the user, like `make_save_report(store, owner)` in `tools.py`; a `{user-preferences}` placeholder in `prompts/system.md`, filled before each model call by dynamic-prompt middleware (like `{user-brands}`, which is filled once today in `app.py`).

If the store is down, the question continues with no preferences. The nightly learning job reads conversations from LangSmith traces; it only writes `status = suggested` rows.

See also Golden Knowledge, report queries, tool factory, middleware, architecture.